VIETNAM GENERAL CONFEDERATION OF LABOUR

TON DUC THANG UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY



HOANG DINH QUY VU – 252805008

Causal Inference and Deep Graphical Representations for Educational Policy Intervention

A Probabilistic Graphical Model Study on UCI and Kaggle Benchmarks

COURSE RESEARCH REPORT

PROBABILISTIC GRAPHICAL MODELS (IT701100)

COMPUTER SCIENCE

Instructor

Nguyen Chi Thien, Ph.D.

HO CHI MINH CITY, 2026

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With heartfelt appreciation, I wish Dr. Nguyen Chi Thien good health, happiness, and continued success in his research and teaching endeavors.

Ho Chi Minh City, November 2026

Author

(Sign and print full name)

Hoang Dinh Quy Vu

DECLARATION

I hereby declare that this course research monograph on *Probabilistic Graphical Models (IT701100)* was carried out independently by myself under the academic guidance and supervision of Dr. Nguyen Chi Thien; and that the empirical work and results contained in it are original and have not been submitted previously for any other degree or course credit. All experimental implementations, data splits, graphical parameterizations, and counterfactual evaluations presented in this report represent my own work.

All external datasets, algorithms, and cited literature—specifically the UCI Machine Learning Repository ID 697 dataset, the Kaggle Playground S4E6 benchmark, and the foundational methodologies by Judea Pearl, Donald Rubin, Xun Zheng, and Amit Sharma—have been duly acknowledged and explicitly cited in the bibliography.

I take full responsibility for the academic integrity and validity of the research presented in this report.

Ho Chi Minh City, November 2026

Author

(Sign and print full name)

Hoang Dinh Quy Vu

CAUSAL INFERENCE AND DEEP GRAPHICAL REPRESENTATIONS FOR EDUCATIONAL POLICY INTERVENTION

ABSTRACT

Estimating the true causal impact of educational interventions from observational tabular records is a fundamental problem in modern institutional policy design. While traditional Causal Bayesian Networks (CBNs) provide rigorous identification guarantees and transparent probabilistic reasoning through Judea Pearl's do-calculus, discrete conditional probability tables (CPDs) suffer from statistical sparsity and struggle to capture continuous or high-order nonlinear interactions. Conversely, purely correlational deep learning and machine learning algorithms achieve high predictive accuracy but fail to identify counterfactual outcomes due to confounding bias and collider stratification.

In this study, we propose and rigorously benchmark a controlled hybrid framework—**Hybrid Neural-Causal Probabilistic Graphical Models (Hybrid PGM)**—that bridges structural causal discovery, Bayesian graphical modeling, and deep neural parameterization. Focusing on the public educational success benchmark (UCI Machine Learning Repository ID 697, $N = 4,424$, binary cohort $N = 3,630$) and its large-scale counterpart (Kaggle Playground Series S4E6, $N = 76,518$), we formalize a 7-node Primary Domain Directed Acyclic Graph (DAG) with 11 directed edges, separating 24 pre-enrollment baseline covariates from 12 post-enrollment academic progression variables. We design a 2×2 factorial evaluation comparing four paradigms: baseline tabular CBN with BDeu smoothing (B_0), fixed-DAG hybrid neural outcome parameterization via PyTorch Multi-Layer Perceptrons (H_1), continuous optimization-based causal discovery via NOTEARS with temporal forbidden masking (H_2), and fully neural graphical discovery-parameterization (H_3).

Across five paired random seeds under a 70/15/15 train/validation/test split,

our empirical results establish three central findings: (1) Granting scholarships ($T = 1$) yields a statistically significant, positive causal average treatment effect on degree graduation ($ATE = +22.3\% \sim +23.7\%$, $p < 10^{-15}$), confirmed by DoWhy placebo and data-subset refutations. (2) Neural CPD parameterization (H_1, H_3) markedly improves predictive probability calibration (lowering Brier score from 0.1719 to 0.1667) and discriminatory ranking (AUROC = 0.8053), while preserving structural causal invariance ($ATE \approx +0.23$). (3) Causal mediation analysis demonstrates that 78.5% of the scholarship effect operates via direct socioeconomic relief ($NDE = +16.96\%$), while 21.5% is mediated through first-semester academic momentum ($NIE = +4.65\%$). Furthermore, Cinelli-Hazlett sensitivity analysis confirms robust unconfoundedness ($RV_{q=1} = 19.09\%$), and budget-constrained counterfactual policy learning reveals that prioritizing scholarship distribution based on Conditional Average Treatment Effects (CATE) produces +164 additional university graduates annually at zero incremental fiscal cost. Cross-dataset testing on 61,578 synthetic students validates that the learned graphical causal mechanisms remain invariant under distribution shifts ($ATE = +0.2143$, AUROC = 0.8537).

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LIST OF ABBREVIATIONS

AIPW	Augmented Inverse Probability Weighting (Doubly Robust)
ATE	Average Treatment Effect
AUROC	Area Under the Receiver Operating Characteristic Curve
BDeu	Bayesian Dirichlet equivalent uniform prior
BN	Bayesian Network
CATE	Conditional Average Treatment Effect
CBN	Causal Bayesian Network
CI	Confidence Interval
CPD	Conditional Probability Distribution
DAG	Directed Acyclic Graph
DL	Deep Learning
DoWhy	End-to-End Library for Causal Inference
ESS	Equivalent Sample Size
GCM	Graphical Causal Model
IPW	Inverse Probability Weighting
MLE	Maximum Likelihood Estimation
MLP	Multi-Layer Perceptron
NDE	Natural Direct Effect
NIE	Natural Indirect Effect
NOTEARS	Non-combinatorial Optimization for Tabular Estimations of Acyclic Relations

OOD	Out-of-Distribution
PGM	Probabilistic Graphical Model
RCM	Rubin Causal Model (Potential Outcomes Framework)
RCT	Randomized Controlled Trial
ROC	Receiver Operating Characteristic
RV	Robustness Value (Cinelli-Hazlett Sensitivity Metric)
SCM	Structural Causal Model
TE	Total Effect
UCI	University of California, Irvine Machine Learning Repository

1 INTRODUCTION

1.1 Reason for Choosing the Topic

In higher education administration, student dropout represents a severe socioeconomic crisis that impacts institutional sustainability, wastes public educational funding, and diminishes the long-term career trajectories of young adults (Martins et al., 2021). Consequently, universities and government agencies worldwide allocate substantial fiscal resources to tuition subsidies, merit-based grants, and need-based scholarships. However, a persistent and unresolved policy dilemma lies at the core of educational decision-making: *Does awarding a scholarship ($T = 1$) causally increase a student's probability of graduating ($Y = 1$), or is the observed correlation merely a spurious statistical artifact resulting from positive self-selection and confounding background advantages?*

Standard predictive machine learning models—such as Gradient Boosted Decision Trees, Random Forests, and deep neural classifiers—excel at predicting academic outcomes by estimating the observational conditional expectation $P(Y = 1 \mid T = t, \mathbf{X} = \mathbf{x})$. Nevertheless, as rigorously established by Pearl (2009) and Hern'an and Robins (2020), statistical association does not imply causation:

$$P(Y = 1 \mid T = t) \neq P(Y = 1 \mid do(T = t)) \tag{1.1}$$

In historical observational data, students who receive scholarships frequently differ systematically from those who do not: they often possess superior baseline entrance examination scores (Z_2), come from households with higher parental education levels (Z_3), and enjoy greater baseline financial stability (Z_5). Comparing the raw graduation rates of scholarship recipients against non-recipients inevitably entangles the true causal effect of financial support with pre-existing selection biases. Naively relying on associational predictive models to inform policy interventions risks committing catastrophic administrative errors, such as misallocating scarce financial resources to

students who would have graduated regardless of institutional intervention.

To overcome these structural limitations, the framework of **Probabilistic Graphical Models (PGMs)**—specifically **Causal Bayesian Networks (CBNs)** and **Structural Causal Models (SCMs)**—provides an expressive mathematical foundation for formal identification and counterfactual reasoning (Ankan & Panda, 2015; Pearl, 2009). Through Judea Pearl's *do*-calculus and graph surgery, CBNs allow researchers to mathematically sever the incoming backdoor confounding paths into the treatment node T, enabling unbiased estimation of the interventional distribution:

$$P(Y = 1 \mid do(T = t)) = \sum_{\mathbf{z}} P(Y = 1 \mid T = t, \mathbf{Z} = \mathbf{z}) P(\mathbf{Z} = \mathbf{z}) \tag{1.2}$$

Despite their mathematical elegance, traditional discrete Bayesian networks rely on Conditional Probability Distributions (CPDs) represented as tabular lookup structures. When conditioning on multiple continuous or high-cardinality demographic confounders, tabular CPDs suffer from extreme parameter explosion and zero-probability sparse cell failures under finite sample sizes (Ankan & Panda, 2015). Conversely, modern Deep Learning (DL) models offer flexible universal function approximation and superior continuous density estimation, yet lack inherent causal semantics, making them vulnerable to learning spurious shortcuts and post-treatment collider stratification biases (Sharma & Kiciman, 2020; Shi et al., 2019).

Therefore, exploring a **Hybrid Neural-Causal Graphical Architecture**—one that unites the formal graph-theoretic identification of Bayesian networks with the nonlinear parameterization and continuous optimization of deep neural networks—represents a vital and timely scientific pursuit.

The scientific and practical significance of this study is threefold:

- **Scientific Contribution:** It establishes an end-to-end, controlled comparative benchmark between classical discrete Bayesian networks and hybrid neural architectures (B_0, H_1, H_2, H_3), investigating whether continuous structural discovery (NOTEARS) and neural outcome parameterization (PyTorch MLP) can

enhance predictive calibration without distorting causal interventional invariance.

- **Methodological Rigor:** It implements multi-stage empirical refutations, including DoWhy placebo and subset permutation tests (Sharma & Kiciman, 2020), Cinelli-Hazlett omitted variable bias sensitivity bounds (Cinelli & Hazlett, 2020), and causal mediation path decomposition ($T \rightarrow M \rightarrow Y$) to isolate whether institutional financial aid operates primarily as direct economic relief or through early academic momentum.

- **Practical Policy Optimization:** It formulates an optimal counterfactual policy learning pipeline under realistic institutional budget constraints, demonstrating that targeting aid based on Conditional Average Treatment Effects (CATE) produces substantial societal dividends—generating +164 additional university graduates annually at zero incremental budget overhead.

The primary beneficiaries of this research include university leadership, academic affairs divisions, student financial aid committees, and educational data scientists seeking defensible, causally grounded artificial intelligence systems for institutional decision support.

1.2 Research Objectives

1.2.1 General Objective

The overarching objective of this research is to design, implement, and rigorously benchmark a principled **Hybrid Neural-Causal Probabilistic Graphical Modeling Framework** for educational policy intervention. By integrating structural causal discovery, Bayesian network inference, and deep neural parameterization, this study aims to accurately estimate the true causal treatment effect of scholarship allocation on degree graduation rates, overcome observational confounding and sample sparsity, and establish an optimal counterfactual policy learning mechanism under institutional capacity constraints.

1.2.2 Specific Objectives

To achieve the general objective, the investigation is structured around four concrete, measurable research milestones:

1. **Causal Problem Formulation and Data Auditing (Objective 1):** Formulate a mathematically defensible causal query on the benchmark educational cohort (UCI Machine Learning Repository ID 697, $N = 4,424$; binary cohort $N = 3,630$). Implement rigorous temporal auditing to isolate pre-treatment confounders ($Z_1..Z_5$) from post-treatment progression variables, eliminate selection bias by filtering intermediate cohorts ('Enrolled'), establish non-zero positivity overlap across demographic strata, and institute an invariant 5-paired random seed splitting protocol (70% Train / 15% Validation / 15% Test).

2. **Controlled 2×2 Factorial Model Implementation (Objective 2):** Design and implement four distinct graphical and parameterization architectures:

    - **Baseline CBN (B_0):** A 7-node, 11-edge Primary Domain Directed Acyclic Graph (DAG) with tabular Bayesian Dirichlet equivalent uniform (BDeu) prior smoothing ($ESS = 5$).

- **Hybrid Neural CPD (H_1):** The fixed domain DAG coupled with a deep PyTorch Multi-Layer Perceptron (MLP) parameterizing the outcome node $P(Y \mid \text{Parents}(Y))$, enforcing strict causal Markov compatibility.
- **Neural Causal Discovery (H_2):** Continuous optimization-based structure learning via NOTEARS (Zheng et al., 2018) constrained by a strictly enforced Temporal Forbidden Mask M ($W_{\text{eff}} = M \odot W$) to eliminate reverse-time causality.
- **Fully Neural Hybrid (H_3):** Uniting NOTEARS-discovered topologies with deep neural outcome CPDs.

Develop a vectorized $O(1)$ Tensor CPD query engine to achieve high-throughput counterfactual inference.

3. **Dual-Criterion Evaluation and Causal Refutation (Objective 3):** Conduct a standardized comparative evaluation across 5 paired runs, strictly separating held-out predictive scoring on Y_{test} (AUROC, Macro-F1, Brier Score) from standardized counterfactual interventional estimation $P(Y = 1 \mid do(T = t))$ on the test reference population C_{test} without touching Y_{test}. Validate the robustness of the causal estimates through DoWhy placebo treatment and data-subset refutations (Sharma & Kiciman, 2020), paired run spread analysis, and Cinelli-Hazlett unobserved confounding sensitivity metrics ($RV_{q=1}$) (Cinelli & Hazlett, 2020).

4. **Advanced Policy Learning, Mediation, and Out-of-Distribution Transfer (Objective 4):** Decompose the estimated total effect into Natural Direct Effects (NDE) and Natural Indirect Effects (NIE) through first-semester academic momentum ($T \rightarrow M \rightarrow Y$). Quantify heterogeneous treatment effects (CATE) across student demographic subgroups (age, debt status, living conditions). Derive an optimal budget-constrained scholarship allocation policy, and validate the out-of-distribution (OOD) generalizability and structural invariance of the

learned causal network on the large-scale synthetic benchmark (Kaggle S4E6, $N = 76,518$).

1.3 Research Questions and Hypotheses

1.3.1 Research Questions

This investigation is structured around one overarching question and four specific, empirically verifiable research questions:

> **RQ0 (Overarching Research Question):** Does financial scholarship provision (T) exert a genuine causal effect in promoting degree completion (Y) among undergraduate students after adjusting for all observable pre-enrollment confounders, through what structural mechanisms does this effect propagate, and can hybrid neural-causal graphical models enhance predictive probability calibration without violating interventional invariance?

This foundational question decomposes into four targeted inquiries:

- **Research Question 1 (RQ1 – Causal Identifiability and Effect Magnitude):** Under the Backdoor adjustment criteria, what is the true Average Treatment Effect (ATE) of awarding a scholarship on the probability of university graduation, and does this estimated effect withstand formal refutation checks (Placebo Treatment and Data-Subset refuters)?

- **Research Question 2 (RQ2 – Neural Parameterization vs. Causal Invariance):** Does substituting discrete tabular CPDs with deep neural Multi-Layer Perceptrons (H_1, H_3) improve held-out predictive accuracy and probability calibration (AUROC, Macro-F1, Brier Score), and does this nonlinear parameterization preserve the numerical stability of the standardized causal interventional query $P(Y \mid do(T))$?

- **Research Question 3 (RQ3 – Structural Mechanisms and Effect Heterogeneity):** What proportion of the scholarship's causal effect operates as a direct socioeconomic shock absorber (Natural Direct Effect - NDE) versus an academic performance mediator (Natural Indirect Effect - NIE via first-semester credit accumulation), and how does the treatment effect vary across student subgroups defined by age, debt status, and living arrangements?

- **Research Question 4 (RQ4 – Robustness and Policy Transferability):** How vulnerable is the causal conclusion to potential unobserved confounders (measured by the Cinelli-Hazlett Robustness Value RV), what graduation dividend can be unlocked via CATE-optimal policy learning under strict capacity constraints, and does a causal graphical model trained on small-scale institutional data generalize out-of-distribution to large-scale synthetic populations?

1.3.2 Research Hypotheses

To provide rigorous falsifiability, we formulate four empirically testable hypotheses:

- **Hypothesis 1 (H1 – Positive and Refutable Causal Treatment Effect):** Awarding a scholarship produces a strictly positive and statistically significant Average Treatment Effect on graduation ($\widehat{\text{ATE}} > +0.20$, $p < 10^{-5}$). When the treatment assignment is randomly permuted in a Placebo refutation test, the estimated treatment effect collapses to zero ($\widehat{\text{ATE}}_{\text{placebo}} \approx 0.000$, $p > 0.05$), confirming that the observed benefit is causal rather than an artifact of baseline selection.

- **Hypothesis 2 (H2 – Neural Calibration Gain with Structural Invariance):** Incorporating deep neural Multi-Layer Perceptrons for outcome probability estimation (H_1) achieves a lower Brier Score ($\Delta_{\text{Brier}} < 0$) and a higher AUROC ($\Delta_{\text{AUROC}} > 0$) compared to the discrete tabular baseline (B_0). Concurrently, under invariant causal DAG structures, the interventional estimands of B_0 and

H_1 converge within a narrow paired spread ($|\widehat{\text{ATE}}_{H_1} - \widehat{\text{ATE}}_{B_0}| < 0.015$), proving that neural parameterization enhances calibration without distorting causal identification.

- **Hypothesis 3 (H3 – Direct Economic Relief Dominance and Age Disparity):** Causal mediation analysis reveals that the Natural Direct Effect (NDE) accounts for the overwhelming majority ($\geq 75\%$) of the total causal effect, operating as a financial safety net that relieves students from full-time employment pressures. Furthermore, Conditional Average Treatment Effects (CATE) exhibit substantial heterogeneity, with older, non-traditional students (> 20 years of age) experiencing treatment benefits at least 1.5× greater than standard-age entrants.

- **Hypothesis 4 (H4 – Policy Dividend and Out-of-Distribution Invariance):** Under historical institutional budget constraints (capping scholarship capacity at 26.7‰), allocating awards via a CATE-optimal decision rule improves overall institutional graduation rates by at least +4.0 percentage points (> 150 additional graduates per cohort) compared to the historical baseline without requiring extra financial capital. Moreover, when transferred out-of-distribution to the 76,518-instance Kaggle benchmark, the Causal Bayesian Network sustains robust predictive discrimination (AUROC > 0.85) and causal stability without performance collapse.

1.4 Research Subjects and Scope

1.4.1 Research Subjects

The subjects of this research are categorized into two interrelated dimensions:

- **Methodological and Algorithmic Subject:** The formulation, implementation, and empirical validation of Probabilistic Graphical Models for causal inference. This includes:

1. Causal Bayesian Networks (CBNs) and Judea Pearl's structural *do*-calculus (Pearl, 2009).
2. Continuous DAG structure discovery via the NOTEARS augmented Lagrangian optimization algorithm (Zheng et al., 2018).
3. Hybrid deep neural outcome parameterization (PyTorch Multi-Layer Perceptrons) constrained by causal Markov factorizations.
4. Formal causal refutation and sensitivity verification mechanisms implemented via the DoWhy framework (Sharma & Kiciman, 2020) and the Cinelli-Hazlett omitted variable bias algebra (Cinelli & Hazlett, 2020).
5. High-throughput vectorized tensor query engines for multi-seed counterfactual inference.

- **Empirical Data Subject:** Higher education administrative records measuring student sociodemographic backgrounds, secondary school qualifications, institutional enrollment status, and degree completion trajectories. Specifically, this study utilizes:

  1. The benchmark *Predict Students' Dropout and Academic Success* dataset (UCI Machine Learning Repository ID 697) (Martins et al., 2021), comprising $N = 4,424$ undergraduate students tracked across 36 original attributes at the Polytechnic Institute of Portalegre, Portugal.
  2. The large-scale companion benchmark from the Kaggle Playground Series Season 4 Episode 6 ($N = 76,518$) (Kaggle, 2024), generated via Conditional Tabular Generative Adversarial Networks (CTGAN) based on the original Portuguese cohort, utilized for statistical power scaling and out-of-distribution (OOD) causal transfer validation.

1.4.2 Research Scope and Boundaries

To ensure analytical rigor and prevent methodological bias, the scope of this study is strictly bounded by the following operational principles:

1. **Causal Horizon and Eligibility Filtering:** The primary causal estimand evaluates the long-term impact of scholarship aid awarded at matriculation (t_1) on final degree graduation (t_2). Consequently, students classified as 'Enrolled' at the conclusion of the survey (794 students in UCI 697; 14,940 students in Kaggle S4E6) represent uncompleted trajectories and are filtered out of the core binary evaluation cohort, establishing a cleanly identified binary target space ($Y \in \{0 : \text{Dropout}, 1 : \text{Graduate}\}$).

2. **Temporal Separation and No-Leakage Principle:** A foundational tenet of causal inference is that pre-treatment adjustment sets must strictly precede the intervention in time ($t_0 \prec t_1$). Consequently, all 12 academic performance variables recorded during the first and second semesters (e.g., enrolled units, approved units, examination grades, and grade point averages) are strictly forbidden from entering the baseline backdoor adjustment set $\mathbf{Z}$. Conditioning on post-treatment variables would induce severe collider stratification bias (M-bias) and block legitimate mediation pathways.

3. **Baseline Confounder Space:** The conditioning set $\mathbf{Z}$ is restricted to five verified pre-enrollment attributes: age at enrollment (Z_1), entrance qualification score (Z_2), maternal educational attainment (Z_3), debt status at enrollment (Z_4), and tuition fee payment status (Z_5). These attributes form a sufficient adjustment set blocking all known backdoor paths between financial assistance (T) and graduation (Y).

4. **Evaluation Cohort Standardization:** Following the strict evaluation protocol mandated by the course specification (Slide 75), held-out predictive metrics

(AUROC, Macro-F1, Brier Score) are evaluated against ground-truth labels Y_{test}. Conversely, interventional risks $P(Y = 1 \mid do(T = t))$ and the Average Treatment Effect (ATE) are standardized across the covariate profile distribution of the held-out reference cohort C_{test}, completely uncoupled from Y_{test}.

5. **Policy Context and Budget Boundaries:** Counterfactual policy learning simulations are constrained to realistic institutional operating limits, holding total scholarship disbursement capacity to the historical institutional ceiling of 26.7% of incoming enrollments.

1.5 Contributions of the Study

This investigation delivers six substantive methodological, technical, and applied contributions to the domains of Probabilistic Graphical Models, Causal Machine Learning, and Educational Policy Informatics:

1. **A Controlled 2×2 Hybrid Neural-Causal Benchmarking Framework:** We design and formalize an end-to-end experimental protocol that systematically crosses structural graph topologies (expert-defined Domain DAG vs. NOTEARS continuous discovery) with probability parameterization families (tabular BDeu Dirichlet smoothing vs. deep PyTorch Multi-Layer Perceptrons). This isolates the exact contribution of deep neural representations within formal causal DAGs.

2. **High-Throughput Vectorized $O(1)$ Tensor CPD Query Engine:** To eliminate the severe computational bottleneck of iterative variable elimination algorithms in large-scale counterfactual queries (which required > 250 seconds for multi-seed bootstrap runs), we engineer a vectorized tensor lookup engine directly indexing multidimensional NumPy arrays (`cpd_y.values`). This reduces counterfactual evaluation runtime from 5.5 seconds per test split to 0.0008 seconds—achieving a $> 3,000\times$ acceleration while preserving exact floating-point precision.

3. **Empirical Proof of Structural Causal Invariance:** We demonstrate across five paired random seeds that while deep neural parameterization (H_1, H_3) markedly improves probability calibration (lowering Brier score from 0.1719 to 0.1667) and ranking capacity (raising AUROC from 0.7955 to 0.8053), the estimated Average Treatment Effect remains rigorously invariant ($\widehat{\text{ATE}} = +0.2221 \sim +0.2370$). This empirically proves that when the underlying causal DAG is structurally sound, deep learning enhancements do not corrupt causal estimands.

4. **Decomposition of Causal Transmission Channels (Mediation Analysis):** We quantify for the first time on the UCI ID 697 benchmark that the scholarship's +21.61% total causal effect decomposes into a 78.5% **Natural Direct Effect** ($NDE = +16.96\%$) and a 21.5% **Natural Indirect Effect** ($NIE = +4.65\%$) transmitted through first-semester credit completion ($T \rightarrow M \rightarrow Y$). This reveals that financial aid acts predominantly as an economic survival buffer rather than solely an academic performance catalyst.

5. **Actionable Budget-Constrained Policy Learning (+164 Graduates/Year at $0 Fiscal Cost):** By ranking incoming students by their model-derived Conditional Average Treatment Effect ($\widehat{\text{CATE}}$), we construct an optimal policy allocation curve under institutional budget constraints. At the historical 26.7% capacity limit, the CATE-optimal policy elevates the institution-wide graduation rate from 60.85% to 65.37% (+4.52%), securing **+164 additional degree completions per cohort** without requiring any budgetary expansion.

6. **Empirical Proof of Out-of-Distribution Invariance on 76k Instances:** We demonstrate that a Causal Bayesian Network trained exclusively on 3,630 authentic university students generalizes seamlessly when deployed counterfactually on 61,578 synthetic students from Kaggle S4E6, preserving both its causal effect magnitude ($ATE = +0.2143$) and probabilistic discrimination (AUROC =

0.8537). This establishes that causal DAG representations avoid the out-of-distribution performance collapse typical of purely associational deep learning models.

1.6 Report Structure

The remainder of this research monograph is organized into five subsequent chapters:

- **Chapter 2 (Theoretical Foundations and Literature Review):** Surveys the mathematical principles of Probabilistic Graphical Models, Directed Acyclic Graphs, and Pearl's structural causal hierarchy (L_1 association, L_2 intervention, L_3 counterfactuals). Reviews continuous causal discovery algorithms (NOTEARS, DAG-GNN), hybrid neural parameterizations, econometric semi-parametric estimators (IPW, AIPW), and the existing literature on higher education retention modeling.

- **Chapter 3 (Methodology and System Architecture):** Formulates the 7-node Primary Domain DAG, mathematically specifies the 2×2 factorial configurations (B_0, H_1, H_2, H_3), and details the architecture of the PyTorch neural CPD and the Temporal Forbidden Mask (M). Outlines the $O(1)$ vectorized tensor query engine and formalizes the DoWhy refutation suite and Cinelli-Hazlett sensitivity algebra.

- **Chapter 4 (Empirical Evaluation and Core Benchmark Results):** Presents the standardized results across 5 paired experimental runs on the UCI ID 697 dataset: Table A (Held-Out Predictive Performance), Table B (Standardized Causal Interventions and ATE), and Table C (DoWhy Refutations, Paired Spread, and Runtime). Details the ablation studies (A1: Dropout regularizer removal, A2: Forbidden mask ablation and reverse causality failure, A3: Maximum likelihood vs. BDeu smoothing).

- **Chapter 5 (Advanced Causal Extensions and Policy Optimization):** Investigates multi-dimensional academic outcomes (GPA and credit velocity), computes causal mediation path decompositions (NDE vs. NIE), explores subgroup treatment heterogeneity (CATE), and presents the Cinelli-Hazlett contour robustness analysis. Formalizes the budget-constrained policy learning curve and details the large-scale 76,518-instance Kaggle S4E6 out-of-distribution transferability analysis.

- **Chapter 6 (Conclusions, Limitations, and Future Directions):** Synthesizes the key theoretical and institutional takeaways, frankly discusses model failure modes and unmeasured confounding boundaries, and charts future research directions combining graph neural diffusion with dynamic continuous-time educational interventions.

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